

## Discussion 5: Matrix multiplication, Taylor series, and velocity addition revisited – **Solutions**

Names:

Section:

This week we introduced two important math methods concepts: matrix multiplication and Taylor series. In this discussion you'll get some practice putting these concepts to use, and along the way you'll link up with the physics by applying these concepts to the context of Lorentz and Galilean transformations.

### Part 1: Matrix multiplication

1.1. Consider the following  $3 \times 3$  matrices:

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix}, \quad R_z = \begin{pmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (1)$$

which (as you will see next week, or you can check for yourself now) represent active counterclockwise rotations by angles  $\alpha$  about the  $x$ -axis and  $\beta$  about the  $z$ -axis, respectively. Compute the matrix product  $R_x R_z$ , and also the matrix product in the reverse order,  $R_z R_x$ . Do you get the same answer? Draw a sketch (or take pictures with your phone) of an object being rotated by these matrices to justify your reasoning. Feel free to choose whichever angles  $\alpha$  and  $\beta$  make the visualization easiest.

$$R_x R_z = \begin{pmatrix} \cos \beta & -\sin \beta & 0 \\ \cos \alpha \sin \beta & \cos \alpha \cos \beta & -\sin \alpha \\ \sin \alpha \sin \beta & \cos \beta \sin \alpha & \cos \alpha \end{pmatrix}, \quad R_z R_x = \begin{pmatrix} \cos \beta & -\cos \alpha \sin \beta & \sin \alpha \sin \beta \\ \sin \beta & \cos \alpha \cos \beta & -\cos \beta \sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{pmatrix},$$

which are manifestly different matrices. For example, taking  $\alpha = \beta = \pi/2$  gives

$$R_x R_z = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix} \quad \text{and} \quad R_z R_x = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

If you put your iPhone flat on the table with the screen facing up (i.e. the normal is in the  $\hat{z}$  direction) and aligned with the  $+\hat{x}$  direction, using  $\alpha = \beta = \pi/2$ ,  $R_x R_z$  (a rotation about the  $z$ -axis, followed by a rotation about the  $x$ -axis) will result in your phone upside-down with the normal to the screen along  $\hat{y}$ , but  $R_z R_x$  results in your phone on its side with the normal pointed along  $-\hat{x}$ .

1.2. Recall the Galilean transformation matrices from Discussion 3:

$$M = \begin{pmatrix} 1 & 0 \\ v & 1 \end{pmatrix}, \quad M' = \begin{pmatrix} 1 & 0 \\ -v & 1 \end{pmatrix}. \quad (2)$$

Compute  $MM'$  and  $M'M$  by matrix multiplication. You should find that both are equal to

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (3)$$

the  $2 \times 2$  identity matrix.

$$\begin{aligned} MM' &= \begin{pmatrix} 1(1) + 0(-v) & 1(0) + 0(1) \\ v(1) + 1(-v) & v(0) + 1(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ M'M &= \begin{pmatrix} 1(1) + 0(v) & 1(0) + 0(1) \\ -v(1) + 1(v) & -v(0) + 1(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

1.3. Multiply two Galilean matrices for different relative velocities,

$$M_v = \begin{pmatrix} 1 & 0 \\ v & 1 \end{pmatrix}, \quad M_w = \begin{pmatrix} 1 & 0 \\ w & 1 \end{pmatrix}. \quad (4)$$

Does matrix multiplication commute (i.e. is  $M_v M_w$  equal to  $M_w M_v$ ) in this instance? Explain how this calculation yields the Galilean velocity addition rule.

$M_v M_w = \begin{pmatrix} 1(1) + 0(w) & 1(0) + 0(1) \\ v(1) + 1(w) & v(0) + 1(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ v+w & 1 \end{pmatrix}$ .  $M_w M_v$  gives the same result, so matrix multiplication commutes in this case. The Galilean velocity rule follows because  $M_w$  can be seen as a Galilean transformation from rest to a frame at velocity  $w$ , and  $M_v$  transforms to a frame with relative velocity  $v$  with respect to  $w$ , so the combined Galilean transformation has velocity  $v + w$  with respect to the rest frame.

1.4. Multiply the Lorentz transformation matrix and its inverse,

$$\Lambda = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix}, \quad \Lambda' = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}. \quad (5)$$

and show that  $\Lambda\Lambda' = \Lambda'\Lambda = I$ .

$\Lambda\Lambda' = \Lambda'\Lambda = \begin{pmatrix} \gamma^2 - \gamma^2\beta^2 & 0 \\ 0 & -\gamma^2\beta^2 + \gamma^2 \end{pmatrix}$ . Using  $\gamma^2 = \frac{1}{1-\beta^2}$ , the diagonal elements simplify to 1, so the matrix multiplication gives the identity matrix  $I$ .

## Part 2: Taylor series

Taylor series are more than a neat calculus trick: they are perhaps the *most important tool* for computing approximations in physics when some parameter is small but nonzero. Recall the definition of a Taylor series from lecture:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n. \quad (6)$$

The most common situation is when  $a = 0$ , in which case  $f(x)$  can be approximated near  $x = 0$  as  $f(x) \approx f(0) + xf'(0)$ . The next most common situation is when  $a = 1$ , which is useful when  $f(x)$  diverges at  $x = 0$  or otherwise behaves badly.

2.1. Use the first two terms of a Taylor series to compute the approximate values of  $\sqrt{405}$  and  $(1003)^{1/3}$ , *without using a calculator!* Note that you will have to manipulate these expressions to match the form of the Taylor series of  $(1+x)^\alpha$  as given in lecture.

$$\sqrt{405} = \sqrt{400} \sqrt{1 + \frac{5}{400}} \approx 20 \left(1 + \frac{5}{2 \cdot 400}\right) = 20 + \frac{1}{8} \approx 20.125.$$

$$(1000)^{1/3} = (1000)^{1/3} \left(1 + \frac{3}{1000}\right)^{1/3} \approx 10 \left(1 + \frac{3}{3 \cdot 1000}\right) \approx 10.01.$$

- 2.2. Perform a Taylor expansion of each component of the Lorentz transformation matrix  $\Lambda$  in (5), taking the small quantity to be  $\beta = v/c$ . Keep only the terms to leading order in  $\beta$  (i.e. ignore anything proportional to  $\beta^2$  or a higher power of  $\beta$ ). Compare your result to the Galilean matrix  $M$  in (2) from Part 1. What is different?<sup>1</sup>

$\Lambda = \begin{pmatrix} (1 - \beta^2)^{-1/2} & \beta(1 - \beta^2)^{-1/2} \\ \beta(1 - \beta^2)^{-1/2} & (1 - \beta^2)^{-1/2} \end{pmatrix} = \begin{pmatrix} 1 + \mathcal{O}(\beta^2) & \beta + \mathcal{O}(\beta^3) \\ \beta + \mathcal{O}(\beta^3) & 1 + \mathcal{O}(\beta^2) \end{pmatrix} \approx \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix}$ . This is almost the same as  $M$ , but the  $\beta$  in the upper-right entry tells us that the correct transformation is  $t = t' + \beta x'/c$ , rather than universal Newtonian time,  $t = t'$ . In other words, clocks separated by large distances are not synchronized to  $\mathcal{O}(\beta)$  in a moving frame, even for small velocities!

- 2.3. Expand the relativistic velocity addition formula,  $\beta_{12} = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}$ , for small velocities. In other words, perform a Taylor expansion of the denominator in the small quantity  $\beta_1 \beta_2 \ll 1$  and keep only the term linear in  $\beta_1 \beta_2$ . Identify the Galilean velocity addition formula as the term linear in  $\beta_1$  and  $\beta_2$ , and the remainder as the leading relativistic correction.

$\frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} = (\beta_1 + \beta_2)(1 + \beta_1 \beta_2)^{-1} \approx (\beta_1 + \beta_2)(1 - \beta_1 \beta_2) = \beta_1 + \beta_2 - (\beta_1^2 \beta_2 + \beta_2^2 \beta_1)$ . The term  $\beta_1 + \beta_2$  is the Galilean formula, and the term in parentheses is the relativistic correction, which is  $\mathcal{O}(\beta^3)$ .

- 2.4. A practical application of your result from 2.3 is provided by the *Fizeau experiment*, which measured the velocity of light in a refractive medium which was moving with respect to the laboratory. Let the medium move with speed  $v \ll c$  with respect to the lab, and consider a light beam emitted within the medium in the same direction of motion. Let the index of refraction be  $n$  (with  $n > 1$ ), so that light in the medium has speed  $c/n$  with respect to the medium. Use the velocity addition formula (note that this setup is analogous to the ball-thrown-from-a-train example) and a Taylor expansion in  $v/c$  to show that the velocity of the light in the lab frame is approximately

$$v_{lab} \approx \frac{c}{n} + v \left( 1 - \frac{1}{n^2} \right). \quad (7)$$

This formula was derived by Max von Laue in 1907, just two years after Einstein's 1905 paper on special relativity, and explained the results of numerous experiments from the 19th century.

Since  $c/n$  is the speed of the light with respect to the medium, and  $v$  is the speed of the medium with respect to the lab, the speed of the light with respect to the lab is  $v_{lab} = \frac{v + c/n}{1 + \frac{v(c/n)}{c^2}} = \frac{v + c/n}{1 + \frac{v}{nc}}$ .

Since  $v \ll c$  and  $n > 1$ ,  $\frac{v}{nc}$  is a suitable small parameter to expand the denominator:  $v_{lab} \approx (v + \frac{c}{n}) (1 - \frac{v}{nc})$ . Expanding the product and rearranging, we have  $\frac{v_{lab}}{c} = \frac{1}{n} + \frac{v}{c} (1 - \frac{1}{n^2}) - \frac{v^2}{c^2} \frac{1}{n}$ . By writing things in this way, both the LHS and RHS are dimensionless, and we are justified in dropping the last term because  $v^2/c^2 \ll v/c$ . Multiplying by  $c$  gives the desired Eq. (7). Note that Galilean relativity would have given  $v_{lab} = c/n + v$ . This experiment, which was performed in 1851 using water as a medium, already showed the problems with Galilean relativity a decade before Maxwell's equations were written down!

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<sup>1</sup>This result is a bit subtle. In fact, the Galilean transformations do *not* arise from the limit  $v \rightarrow 0$ , but rather  $c \rightarrow \infty$ . In a mathematical sense, these are actually quite different limits. But since  $c$  is not infinite in our world, your result (and *not* the Galilean one) is actually the correct transformation! See Morin 2.7 for more discussion.

## Part 3: Velocity addition revisited

- 3.1. The velocity addition formula is symmetric in  $v_1$  and  $v_2$ . Show that  $\Lambda_{v_1}\Lambda_{v_2}$  gives the same matrix as  $\Lambda_{v_2}\Lambda_{v_1}$  computed in class. In other words, *longitudinal* velocity addition (along the same direction) is symmetric because Lorentz transformations along the same direction commute.

$$\Lambda_{v_1}\Lambda_{v_2} = \begin{pmatrix} \gamma_1 & \gamma_1\beta_1 \\ \gamma_1\beta_1 & \gamma_1 \end{pmatrix} \begin{pmatrix} \gamma_2 & \gamma_2\beta_2 \\ \gamma_2\beta_2 & \gamma_2 \end{pmatrix} = \begin{pmatrix} \gamma_1\gamma_2(1+\beta_1\beta_2) & \gamma_1\gamma_2(\beta_1+\beta_2) \\ \gamma_1\gamma_2(\beta_1+\beta_2) & \gamma_1\gamma_2(1+\beta_1\beta_2) \end{pmatrix} \text{ which is the same as } \Lambda_{v_2}\Lambda_{v_1}.$$

- 3.2. Now consider *transverse* velocity addition: suppose that an object moves in the  $xy$ -plane with velocity  $(u'_x, u'_y)$  with respect to  $S'$ , which is moving in the  $+x$ -direction at speed  $v$  with respect to  $S$ . Find the velocity of the object in  $S$  by identifying its components  $u_x$  and  $u_y$ . To do this, use the fact that the  $\Delta$ 's in the Lorentz transformations can be treated as infinitesimals, and  $u'_x \equiv \Delta x'/\Delta t'$  and  $u'_y \equiv \Delta y'/\Delta t'$ . You want to find  $u_x \equiv \Delta x/\Delta t$  and  $u_y \equiv \Delta y/\Delta t$ . Express your answers in terms of  $u'_x, u'_y, v$ , and  $\gamma_v \equiv 1/\sqrt{1-v^2/c^2}$ , the Lorentz factor between  $S$  and  $S'$ . (Feel free to refer to Morin Sec. 2.2.2).

Writing the Lorentz transformations in terms of  $\Delta$ 's, we have  $\Delta t = \gamma_v \left( \Delta t' + \frac{v\Delta x'}{c^2} \right)$ ,  $\Delta x = \gamma_v(\Delta x' + v\Delta t')$ , and  $\Delta y = \Delta y'$ . First consider the  $y$  velocity:

$$\frac{\Delta y}{\Delta t} = \frac{\Delta y'}{\gamma_v \left( \Delta t' + \frac{v\Delta x'}{c^2} \right)} = \frac{\Delta y'/\Delta t'}{\gamma_v \left( 1 + \frac{v}{c^2} \frac{\Delta x'}{\Delta t'} \right)} = \frac{u'_y}{\gamma_v(1 + u'_x v/c^2)}$$

The  $x$ -velocity is similarly

$$\frac{\Delta x}{\Delta t} = \frac{\gamma_v(\Delta x' + v\Delta t')}{\gamma_v \left( \Delta t' + \frac{v\Delta x'}{c^2} \right)} = \frac{\Delta x'/\Delta t' + v}{1 + (v/c^2)(\Delta x'/\Delta t')} = \frac{u'_x + v}{1 + u'_x v/c^2}.$$

In both calculations, the critical step is dividing both numerator and denominator by  $\Delta t'$  to get things in terms of velocities.

- 3.3. For the special case  $u'_x = 0$ , we can imagine the velocity  $u'_y$  as arising from a Lorentz transformation of a *stationary* object in a new frame  $S''$ , which moves in the  $y$ -direction with respect to  $S'$  at velocity  $u'_y$ . The combined transformation from  $S''$  to  $S'$  and finally to  $S$  is given by the product of Lorentz matrices  $\Lambda_x\Lambda_y$ , with

$$\Lambda_x = \begin{pmatrix} \gamma_v & \gamma_v\beta_v & 0 & 0 \\ \gamma_v\beta_v & \gamma_v & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_y = \begin{pmatrix} \gamma_u & 0 & \gamma_u\beta_u & 0 \\ 0 & 1 & 0 & 0 \\ \gamma_u\beta_u & 0 & \gamma_u & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (8)$$

where  $\gamma_v \equiv 1/\sqrt{1-v^2/c^2}$ ,  $\beta_v \equiv v/c$ ,  $\gamma_u \equiv 1/\sqrt{1-(u'_y)^2/c^2}$ , and  $\beta_u \equiv u'_y/c$ . Perform the matrix multiplication and compute  $\Delta x/\Delta t$  and  $\Delta y/\Delta t$  for  $\Delta x'' = \Delta y'' = \Delta z'' = 0$  (since the object is at rest in  $S''$ ). You should obtain  $u_x = v$  and  $u_y = u'_y/\gamma_v$ , consistent with your results from 3.2 when  $u'_x = 0$ .

$$\Lambda_x \Lambda_y = \begin{pmatrix} \gamma_v \gamma_u & \gamma_v \beta_v & \gamma_v \gamma_u \beta_u & 0 \\ \gamma_v \beta_v \gamma_u & \gamma_v & \gamma_v \beta_v \gamma_u \beta_u & 0 \\ \gamma_u \beta_u & 0 & \gamma_u & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \text{ This is the matrix which transforms } (c\Delta t'', \Delta x'', \Delta y'', \Delta z'')$$

to  $(c\Delta t, \Delta x, \Delta y, \Delta z)$ . Setting  $\Delta x'' = \Delta y'' = \Delta z'' = 0$ , the first column of the matrix lets us immediately read off the coordinates in  $S$ :

$$c\Delta t = \gamma_v \gamma_u (c\Delta t''), \quad \Delta x = \gamma_v \beta_v \gamma_u (c\Delta t''), \quad \Delta y = \gamma_u \beta_u (c\Delta t''), \quad \Delta z = 0.$$

Substituting  $\Delta t'' = \frac{\Delta t}{\gamma_v \gamma_u}$  using the first equation gives  $\Delta x / \Delta t = c\beta_v$ , or  $u_x = v$ , and  $\Delta y / \Delta t = c\beta_u / \gamma_v$ , or  $u_y = u'_y / \gamma_v$ .

- 3.4. Now perform the matrix multiplication from 3.3 in the opposite order: is  $\Lambda_y \Lambda_x$  the same matrix as  $\Lambda_x \Lambda_y$ ? In other words, is transverse velocity addition symmetric? (Hint: if you find *one* element of the matrix that is not equal for the two orderings, you can stop; for the matrices to be equal, *all* entries must be the same.)

The matrices are *not* the same: for example, the second entry in the first column of  $\Lambda_y \Lambda_x$  is  $\gamma_v \gamma_u \beta_u$ , which is not the same as the analogous entry in  $\Lambda_x \Lambda_y$  if  $\beta_u \neq \beta_v$ .

- 3.5. How could we verify whether or not  $\Lambda_x \Lambda_y$  is a Lorentz transformation? (Don't worry if you don't have a complete solution to this problem – the full story will come in the next two weeks!)

In order to do this, we would want to find a velocity vector (which may not be along one of the coordinate axes) such that a boost along that vector yields the matrix  $\Lambda_x \Lambda_y$ . It turns out this is impossible:  $\Lambda_x \Lambda_y$  is not a pure boost, but a combination of a boost and a rotation. We will see in Discussion 7 how this works!